

Pasqal DM

Gras Mathieu
Mercat Hugo

January 13, 2025

1

In this problem the antennas can be mapped as nodes on a graph with vertices between nodes where their distance from each other is less than 8.7 km. In this case, we use the coordinates to the scale $1 = 1\text{km}$.

To encode our problem into this graph we can color the graph using different colors representing the different possible frequencies.

However, a classic undirected graph would not work well in this case because it does not represent well how much two antennas interfere with each other (how much ground is shared between their radius) and it cannot represent well territories shared by more than 2 antennas. Hence an unit disk graph is the best model for this problem.

For this problem, we can map the antennas as disks with a radius of 8.7 in the graph then try to color it with as few colors as possible while also keeping the interference at a minimum.

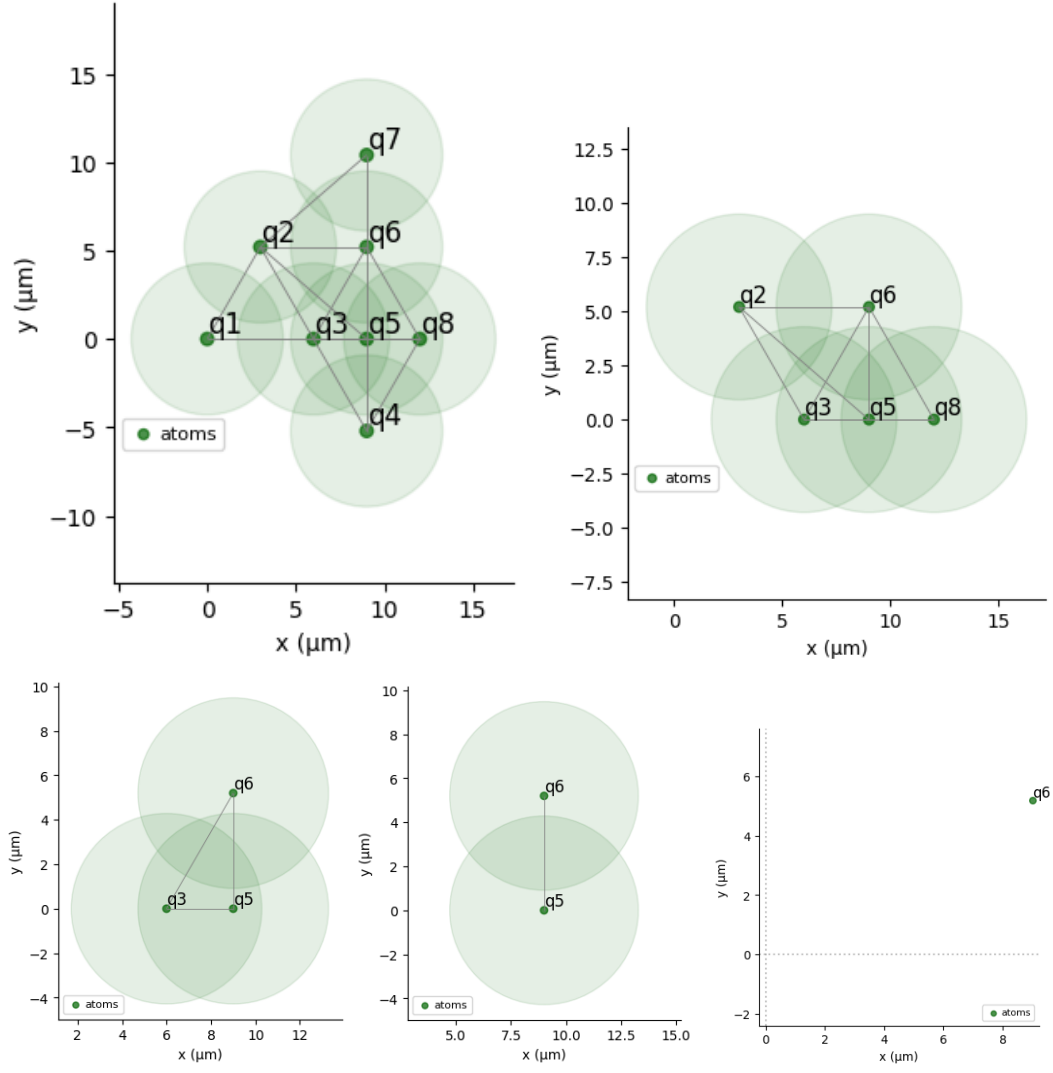
2

For this problem we used an adiabatic solver. We write a function `MIS_solver` that takes a register in the parameter. We first initialize a register with all the qubits at their position.

We set the interference radius to 8.7. Then we apply the `MIS_solver` function to extract an independent set from our register. The qubits set to 1 in the result are colored with the first color.

Then, we recreate another register without the already colored qubit. We apply the `MIS_solver` function and repeat the process until the graph we generate is fully independent.

We began with a version where we needed to iterate manually, but we created a function to iterate automatically until the final condition is met.



As we can see, after each iteration the graph is reduced by the biggest independent graph available until we got one antenna left which is its own biggest independent graph. Hence we got 4 iterations plus the last antenna which gives us 5 different frequencies for our client.